

BIOS AND POST TROUBLESHOOTING THEORY

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

ANS:

BIOS/POST Error Message	Usually Indicates
1. CMOS Checksum Error	CMOS battery failure or incorrect BIOS settings.
2. Keyboard Error / Keyboard Not Found	Keyboard is disconnected or faulty.
3. Memory Error / Memory Test Failure	RAM is loose, incompatible, or faulty.
4. CPU Fan Error	CPU fan is disconnected, blocked, or not working properly.
5. No Boot Device Found	HDD/SSD is not detected or the boot device/order is configured incorrectly.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

ANS:

POST Beep Code Troubleshooting

For a **traditional AMI BIOS**, **3 short beeps** typically indicate a **base-memory (first 64 KB RAM) failure**.

Two troubleshooting steps:

1. **Power off the PC and reseat the RAM** firmly in its slot, then try booting again.
2. **Test one RAM module at a time** or try a known-good compatible RAM module to identify a faulty stick/slot.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

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Steps:

1. Restart the PC.
2. When the PC starts, press **Delete** or **F2** repeatedly.
3. The **BIOS Setup Utility** will open.
4. Go to the **Main** or **System Information** section.

5. Check the **Motherboard Manufacturer** and **BIOS Version**.
6. Take a screenshot of the main BIOS screen.
7. Save the screenshot and paste it into the assignment.

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

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How to resolve:

1. Enter the BIOS setup and check the date, time, and hardware settings.
2. Select **Load Optimized/Default Settings**, then save and restart.
3. If the error appears again, turn off and unplug the PC, then **replace the CMOS battery (usually CR2032)**.
4. Enter BIOS again, set the correct date/time and required settings, save, and restart the PC.